

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO2	Assessment:	Assessment Tool 2 – Scenario-Based Requirement Analysis
Weightage:	20%	Total Marks:	25
CO2 Statement:	<i>Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills</i>		
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Date:	05/08/2026	Duration:	60 minutes

Code of Conduct

I P. Lohith Krishna 192325122(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: P. Lohith Krishna

Annexure A – Scenario-Based Requirement Analysis

Smart Tax Calculation and Financial Planning Management System

1. Study and Analyze the Scenario

- Carefully understand the given problem statement.

Problem Statement

The **Smart Tax Calculation and Financial Planning Management System** is a software application designed to help individuals and small businesses calculate taxes accurately and manage their financial activities efficiently. The system automates tax calculations based on current tax rules, tracks income and expenses, generates financial reports, and provides suggestions for tax savings and financial planning.

Business Domain

- Finance and Tax Management
- Personal Finance
- Financial Planning
- Tax Filing Services

Stakeholders

- Individual Taxpayers
- Small Business Owners
- Chartered Accountants (CAs)
- Financial Advisors

- System Administrator
- Government Tax Department (Indirect Stakeholder)

Existing Challenges

- Manual tax calculations often lead to errors.
- Difficulty in maintaining income and expense records.
- Lack of awareness about tax-saving deductions and investments.
- Time-consuming preparation of tax reports.
- Poor financial planning and budgeting.
- Risk of missing tax filing deadlines.

System Objectives

- Automate tax calculations based on current tax laws.
 - Store and manage users' financial records securely.
 - Generate accurate tax and financial reports.
 - Help users plan budgets and manage expenses.
 - Recommend tax-saving investments and deductions.
 - Send reminders for tax filing and important financial deadlines.
 - Reduce manual effort, improve accuracy, and support better financial decision-making.
- Identify the business domain, stakeholders, existing challenges, and system objectives.

Business Domain

The **Smart Tax Calculation and Financial Planning Management System** belongs to the **Finance and Tax Management** domain. It helps individuals and small businesses manage their income, expenses, tax calculations, budgeting, and financial planning through an automated software solution.

Stakeholders

Primary Stakeholders

- **Individual Taxpayers** – Use the system to calculate taxes and manage personal finances.
- **Small Business Owners** – Maintain business financial records and calculate taxes.
- **Chartered Accountants (CAs)** – Assist clients with tax preparation, verification, and filing.
- **System Administrator** – Manages users, updates tax rules, and maintains the system.

Secondary Stakeholders

- **Financial Advisors** – Provide investment and financial planning advice based on user data.
 - **Government Tax Department** – Receives tax reports and ensures compliance with tax regulations.
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Existing Challenges

- Manual tax calculations are time-consuming and prone to errors.
 - Difficulty in tracking income, expenses, and financial records.
 - Lack of awareness about available tax deductions and tax-saving investments.
 - Delays in generating tax reports and filing tax returns.
 - Poor budgeting and financial planning.
 - Risk of missing tax filing deadlines and penalties.
 - Data stored manually may be lost or difficult to retrieve.
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System Objectives

- Automate tax calculations using the latest tax rules and tax slabs.
- Maintain secure records of users' income, expenses, deductions, and investments.
- Generate accurate tax reports and financial statements.
- Assist users in budgeting and financial planning.
- Recommend tax-saving options and investment opportunities.
- Send reminders for tax filing deadlines and important financial events.
- Improve accuracy, reduce manual effort, and enhance overall financial management.

2. Identify Functional and Non-Functional Requirements

- List all functional requirements required by the proposed system.

Functional Requirements (Points)

1. User Registration – Users can create a new account.
2. User Login – Users can securely log in using their credentials.
3. Password Recovery – Users can reset forgotten passwords.
4. Profile Management – Users can view and update their personal information.
5. Income Management – Add, edit, delete, and view income details.
6. Expense Management – Add, edit, delete, and view expense records.
7. Investment Management – Record investments and tax-saving schemes.
8. Tax Deduction Management – Enter eligible tax deductions.
9. Automatic Tax Calculation – Calculate tax based on income, deductions, and tax slabs.
10. Tax Slab Management – Administrator can update tax rules and tax slabs.
11. Budget Planning – Create and manage monthly or yearly budgets.
12. Financial Planning – Provide savings and investment recommendations.
13. Report Generation – Generate tax, income, expense, and financial reports.

14. Report Download – Download reports in PDF or Excel format.
15. Notifications – Send reminders for tax filing, bill payments, and financial updates.
16. Dashboard – Display income, expenses, savings, and tax summary.
17. Document Upload – Upload receipts, invoices, and tax-related documents.
18. Search and Filter – Search and filter financial records by date, category, or amount.
19. User Management – Administrator can manage users and their permissions.
20. Database Backup and Recovery – Automatically back up and restore system data.
21. Audit Log – Maintain records of user and system activities.
22. Logout – Allow users to securely log out of the system.

- Identify suitable non-functional requirements such as performance, security, reliability, usability, scalability, maintainability, and availability.

Non-Functional Requirements (Points)

Performance

1. The system should respond to user requests within **3 seconds**.
2. It should support multiple users simultaneously without performance degradation.
3. Reports and tax calculations should be generated quickly.

Security

4. Users must log in using secure authentication.
5. Passwords should be encrypted before storage.
6. User financial data must be protected from unauthorized access.
7. Secure communication should be provided using HTTPS/SSL.

Reliability

8. The system should provide **99.9% uptime**.
9. Automatic backup of user data should be performed regularly.
10. The system should recover data after unexpected failures.

Usability

11. The user interface should be simple and easy to navigate.
12. The system should be user-friendly for both beginners and professionals.
13. The application should support mobile and desktop devices.

Scalability

14. The system should support an increasing number of users and financial records.
15. It should be easy to upgrade with new features and tax rules.

Maintainability

16. The software should have a modular design for easy maintenance.
17. System updates and bug fixes should be implemented with minimal downtime.
18. Documentation should be available for developers and administrators.

Availability

19. The system should be available **24×7** for users.
20. Scheduled maintenance should have minimal impact on system availability.

Compatibility

21. The system should work on major web browsers (Chrome, Edge, Firefox, Safari).
22. It should support Windows, Linux, Android, and iOS devices.

Data Integrity

23. The system should maintain accurate and consistent financial records.
24. Duplicate or invalid data entries should be prevented.

3. Identify Actors and Use Cases

- Identify all primary and secondary actors interacting with the system.

Primary Actors

1. **Individual Taxpayer (User)** – Registers, logs in, manages income and expenses, calculates taxes, and views reports.
2. **Small Business Owner** – Manages business financial records, calculates taxes, and generates financial reports.
3. **Chartered Accountant (CA)** – Reviews client financial data, verifies tax calculations, and assists with tax filing.
4. **System Administrator** – Manages users, updates tax rules, monitors the system, and maintains the database.

Secondary Actors

1. **Financial Advisor** – Reviews financial reports and provides investment and tax-saving recommendations.
2. **Government Tax Department** – Receives tax reports and ensures compliance with tax regulations.
3. **Bank/Payment Gateway (Optional)** – Processes online tax payments and financial transactions.
4. **Email/SMS Notification Service** – Sends reminders for tax filing deadlines, bill payments, and system notifications.

- List the major use cases associated with each actor.

1. Individual Taxpayer (User)

- Register a new account
 - Login/Logout
 - Manage profile
 - Add income details
 - Add expense details
 - Enter tax deductions
 - Calculate tax
 - Create budget plans
 - View and download tax reports
 - Receive tax filing reminders
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2. Small Business Owner

- Register/Login
 - Manage business income and expenses
 - Record business investments
 - Calculate business tax
 - Generate financial and tax reports
 - Download reports
 - Track budgets
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3. Chartered Accountant (CA)

- Login
 - View client financial records
 - Verify tax calculations
 - Generate tax reports
 - Assist in tax filing
 - Suggest tax-saving options
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4. System Administrator

- Login
- Manage user accounts
- Update tax slabs and tax rules
- Monitor system activities
- Manage reports
- Perform database backup and recovery
- Maintain audit logs

5. Financial Advisor

- Login
- View financial reports
- Analyze user financial data
- Recommend investment plans
- Suggest tax-saving strategies

6. Government Tax Department

- Receive tax reports
- Verify tax compliance
- Process tax submissions
- Update tax regulations (if integrated)

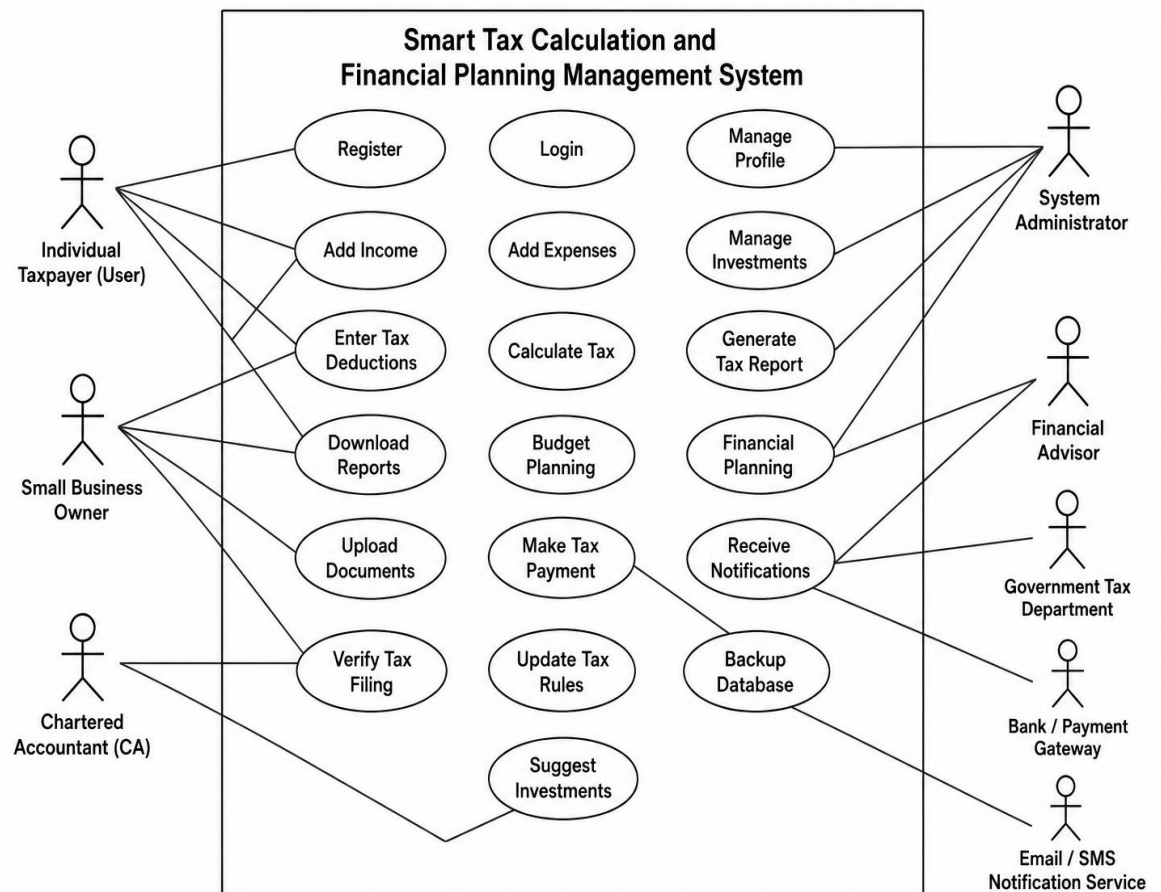
7. Bank/Payment Gateway (Optional)

- Process online tax payments
- Confirm payment status
- Generate payment receipts

8. Email/SMS Notification Service

- Send tax filing reminders
- Send payment confirmations
- Send financial alerts
- Send account-related notifications

- Draw a Use Case Diagram representing the interactions between actors and the system.



4. Prepare the Software Requirement Specification (SRS)

Develop an SRS document containing:

1. Introduction

The **Smart Tax Calculation and Financial Planning Management System** is a software application designed to help individuals and small businesses manage their finances efficiently. It automates tax calculations, tracks income and expenses, generates financial reports, and provides tax-saving recommendations. The system reduces manual effort, improves accuracy, and supports better financial planning.

2. Problem Description

Many taxpayers calculate taxes manually, which often results in errors, delayed tax filing, and poor financial management. Maintaining financial records and identifying tax-saving opportunities is difficult. This system solves these problems by automating tax calculations, organizing financial records, and providing financial planning tools.

3. Scope of the System

The system allows users to:

- Register and log in securely.
 - Manage personal and business profiles.
 - Record income, expenses, investments, and tax deductions.
 - Automatically calculate taxes based on current tax rules.
 - Generate tax and financial reports.
 - Plan budgets and monitor savings.
 - Receive tax filing reminders and notifications.
 - Allow administrators to manage users and update tax rules.
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4. Functional Requirements

- User Registration
 - User Login and Logout
 - Password Recovery
 - Profile Management
 - Income Management
 - Expense Management
 - Investment Management
 - Tax Deduction Management
 - Automatic Tax Calculation
 - Budget Planning
 - Financial Planning
 - Report Generation
 - Report Download (PDF/Excel)
 - Document Upload
 - Notifications and Reminders
 - Search and Filter Records
 - User Management (Admin)
 - Tax Rule Management (Admin)
 - Database Backup and Recovery
 - Audit Log Maintenance
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5. Non-Functional Requirements

Performance

- Response time should be less than 3 seconds.
- Support multiple users simultaneously.

Security

- Secure user authentication.
- Password encryption.
- HTTPS communication.
- Secure storage of financial data.

Reliability

- 99.9% system availability.
- Automatic data backup and recovery.

Usability

- Simple and user-friendly interface.
- Easy navigation.
- Mobile-responsive design.

Scalability

- Support increasing users and financial records.
- Easy integration of new tax rules.

Maintainability

- Modular software design.
- Easy updates and maintenance.

Availability

- System available 24×7 with minimal downtime.

6. Use Case Descriptions

Use Case 1: User Login

Actor: User

Precondition:

- User is registered.

Steps:

1. Enter username/email and password.
2. Click Login.
3. System verifies credentials.
4. Dashboard is displayed.

Postcondition:

- User successfully accesses the system.
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Use Case 2: Calculate Tax

Actor: User

Precondition:

- Income and deduction details are entered.

Steps:

1. Select financial year.
2. Enter deductions.
3. Click Calculate Tax.
4. System computes tax.
5. Display tax summary.

Postcondition:

- Tax calculation report is generated.
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Use Case 3: Generate Financial Report

Actor: User

Steps:

1. Select report type.
2. Choose date range.
3. Click Generate Report.
4. View or download the report.

Postcondition:

- Financial report is available.
-

7. Data Requirements**User Data**

- User ID
- Name
- Email
- Mobile Number
- Password

Financial Data

- Income Details
- Expense Records
- Investment Details
- Tax Deductions
- Budget Information

Tax Data

- Tax Slabs
- Financial Year
- Tax Amount
- Tax Rules

Report Data

- Monthly Reports
 - Annual Reports
 - Tax Reports
 - Financial Summary
-

8. External Interface Requirements

User Interface

- Web-based interface
- Mobile-friendly design
- Dashboard with charts and reports

Hardware Interface

- Computer/Laptop
- Smartphone/Tablet
- Internet Connection

Software Interface

- Web Browser (Chrome, Edge, Firefox, Safari)
- MySQL Database
- Web Server

- Email/SMS Notification Service
 - Payment Gateway (Optional)
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9. System Features

- Secure User Registration and Login
- Income and Expense Tracking
- Automatic Tax Calculation
- Tax Deduction Management
- Budget and Financial Planning
- Investment Suggestions
- Tax Saving Recommendations
- Financial Dashboard
- Report Generation and Download
- Document Upload
- Search and Filter Records
- Email/SMS Notifications
- Admin Panel for User and Tax Rule Management
- Database Backup and Recovery
- Audit Log Management
- 24×7 Secure System Access

5. Identify Assumptions and Constraints

- List the assumptions considered while developing the system.

Assumptions

1. Users provide accurate and complete financial information.
2. Users have a valid account to access the system.
3. A stable internet connection is available while using the application.
4. The system uses the latest government tax rules and tax slabs.
5. Tax rules are updated regularly by the system administrator.
6. Users have basic knowledge of computers or smartphones.
7. Financial data entered by users is genuine and up to date.
8. The database is available and functioning properly.
9. The system is accessed through supported web browsers or mobile devices.
10. Email/SMS services are available for sending notifications and reminders.
11. Regular backups are performed to prevent data loss.
12. Users follow security practices such as keeping login credentials confidential.

- Identify technical, operational, economic, legal, and time-related constraints affecting the project.

Technical Constraints

1. The system requires a stable internet connection.
 2. Compatible with modern web browsers and mobile devices.
 3. Depends on database availability for storing and retrieving data.
 4. Requires regular updates to tax rules and software.
 5. Data backup and recovery mechanisms must be maintained.
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Operational Constraints

1. Users must register before accessing the system.
 2. Only authorized administrators can update tax rules and manage users.
 3. Users are responsible for entering accurate financial information.
 4. The system depends on Email/SMS services for notifications.
 5. Regular maintenance may temporarily interrupt system availability.
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Economic Constraints

1. Limited budget for software development and deployment.
 2. Costs for cloud hosting, database, and server maintenance.
 3. Expenses for third-party services such as Email/SMS and payment gateways.
 4. Ongoing maintenance and software update costs.
 5. Training costs for administrators and support staff.
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Legal Constraints

1. Must comply with government tax laws and regulations.
 2. Must protect user financial data according to data privacy laws.
 3. User information should not be shared without authorization.
 4. Digital records must comply with legal standards for tax documentation.
 5. The system should follow cybersecurity and data protection regulations.
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Time-Related Constraints

1. The project should be completed within the scheduled academic/project timeline.
2. Tax rules must be updated before each new financial year.
3. Reports and tax calculations should be generated within a few seconds.
4. Regular software maintenance should be completed during off-peak hours.
5. System updates should be released on time without affecting users significantly.

6. Validate Requirement Completeness and Consistency

- Verify whether all stakeholder requirements have been addressed.

The following stakeholder requirements have been addressed by the proposed system:

- **✓ Individual Taxpayer:** Can register, log in, manage income and expenses, calculate taxes, generate reports, and receive reminders.
 - **✓ Small Business Owner:** Can manage business financial records, calculate taxes, create budgets, and generate financial reports.
 - **✓ Chartered Accountant (CA):** Can review client records, verify tax calculations, generate tax reports, and assist in tax filing.
 - **✓ Financial Advisor:** Can access financial reports and provide investment and tax-saving recommendations.
 - **✓ System Administrator:** Can manage users, update tax rules, maintain the database, monitor system activities, and perform backups.
 - **✓ Government Tax Department:** Can receive tax reports and ensure compliance with tax regulations (if integrated).
 - **✓ Bank/Payment Gateway:** Supports secure online tax payments and payment confirmation (optional feature).
 - **✓ Email/SMS Notification Service:** Sends tax filing reminders, payment confirmations, and system notifications.
- Check for ambiguity, redundancy, inconsistency, and missing requirements.

1. Ambiguity Check

- All requirements are clearly stated and easy to understand.
- User roles and responsibilities are well defined.
- Functional and non-functional requirements are specific and measurable.

2. Redundancy Check

- Duplicate requirements have been removed.
- Each function is listed only once to avoid repetition.
- Similar features are grouped under appropriate modules.

3. Inconsistency Check

- Functional requirements are consistent with the system objectives.
- User roles do not conflict with each other.
- Security, performance, and usability requirements are compatible with the system design.
- No conflicting business rules or system functions were identified.

4. Missing Requirements Check

The core requirements are covered. However, the following can be added as future enhancements:

- Multi-language support.
 - Mobile application support.
 - Integration with banking and payment systems.
 - AI-based financial planning and tax-saving recommendations.
 - Two-factor authentication (2FA) for enhanced security.
 - Cloud backup and data synchronization.
 - Real-time updates of government tax rules.
 - Chatbot for customer support.
- Suggest improvements to enhance the quality and completeness of the requirement specification.
1. Add **AI-based tax-saving and financial planning recommendations**.
 2. Implement **Two-Factor Authentication (2FA)** to improve account security.
 3. Integrate the system with **government tax portals** for online tax filing.
 4. Provide **real-time tax rule updates** whenever government regulations change.
 5. Develop a **mobile application** for Android and iOS users.
 6. Add **multi-language support** to improve accessibility.
 7. Enable **cloud backup and data synchronization** for secure data storage.
 8. Integrate with **banking and payment gateways** for online tax payments.
 9. Include **data validation** to prevent incorrect or incomplete financial information.
 10. Generate **advanced financial reports and graphical dashboards** for better analysis.
 11. Add **role-based access control** to restrict unauthorized access.
 12. Provide **audit logs** to track user activities and system changes.
 13. Include **automated email and SMS reminders** for tax filing deadlines.
 14. Conduct **regular testing and user feedback sessions** to improve system usability and performance.
 15. Ensure compliance with **data privacy and government tax regulations** to maintain security and legal standards.

Conclusion:

Implementing these improvements will make the **Smart Tax Calculation and Financial Planning Management System** more secure, reliable, user-friendly, and scalable, while ensuring that the Software Requirement Specification (SRS) is complete and of high quality.

Rubrics for Calculation

Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Scenario Understanding and Problem Analysis (30%)	Demonstrates thorough understanding of the scenario, stakeholders,	Demonstrates good understanding with minor omissions.	Basic understanding with limited analysis.	Poor understanding; major aspects missing or incorrect.

	objectives, and business context with clear analysis.			
Functional and Non-Functional Requirement Identification (25%)	Clearly identifies comprehensive, accurate, and well-classified functional and non-functional requirements.	Identifies most requirements with minor classification errors.	Identifies only basic requirements; several omissions.	Requirements are incomplete, inaccurate, or poorly classified.
Actor Identification and Use Case Modelling (20%)	Correctly identifies all relevant actors and use cases; use case diagram is complete, accurate, and follows UML standards.	Most actors and use cases identified; diagram has minor errors.	Limited actors/use cases identified; diagram partially correct.	Actors/use cases missing or incorrect; diagram absent or inaccurate.
Software Requirement Specification (SRS) Documentation (15%)	SRS is well-structured, complete, professionally organized, and includes all required sections.	SRS is organized with minor missing sections or formatting issues.	SRS includes only essential sections with limited detail.	SRS is incomplete, poorly organized, or lacks essential components.
Assumptions, Constraints, and Requirement Validation (10%)	Clearly identifies realistic assumptions and constraints; validates requirements for completeness, consistency, and ambiguity with appropriate justification.	Identifies most assumptions and constraints; validation is adequate.	Limited assumptions/constraints; validation is superficial.	Assumptions, constraints, and validation are missing or incorrect.

